An Explainable Stacking Ensemble Framework with Disagreement-Aware Feature Selection for Loan Approval Prediction

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Abstract*— A loan is money borrowed from a financial institution, which is usually repaid over a period of time with interest. Therefore, accurate approval decisions are crucial in a modern credit system. Although previous studies have shown some useful results, in many cases they have been limited by mediocre performance, poor hyperparameter (HP) tuning, lack of robust cross-validation (CV), inadequate feature selection, limited ablation analysis, and low transparency for practical decision support. To overcome these limitations, this paper proposes an end-to-end framework for loan approval prediction, which combines Disagreement-Aware Feature Selection (DAFS), Random Over-Sampling (ROS), ensemble learning, external validation, and Explainable AI (XAI). After data cleaning, label encoding, scaling, and stratified splitting, DAFS combines Elastic Net stability, permutation importance, and Genetic Algorithm search to select stable and important predictors; at the same time, it imposes penalties for disagreements among different selectors. ROS is applied only to the training data, so that class balancing can be done fairly. The main model is a stacking ensemble of K-Nearest Neighbors (KNN) and Random Forest (RF), which is combined with a Logistic Regression (LR) meta-learner. This method leverages the complementary strengths of instance-based learning and tree-based learning to provide more reliable final decisions. The proposed model achieved 96.93% accuracy on the test set, and further supported the empirical robustness of ablation analysis and external validation, with no evidence of overfitting or underfitting. For interpretability, both global and individual-level decision logics are expressed using SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and permutation importance. As a result, this study combines accuracy, transparency, and trust where more intelligent approval decisions pave the way for fairer financial opportunities.

Keywords—Loan approval prediction, Disagreement-Aware Feature Selection, Stacking ensemble, Random Forest, K-Nearest Neighbors, Explainable AI.

I. INTRODUCTION

The creditworthiness of an applicant is assessed during the loan approval process. Traditional methods often depend on manual review and statistical models. However, these methods are not always scalable for large and complex applicant data. They may also increase biased decisions and default risk [1]. Applicant financial, demographic, and behavioral factors are used to estimate repayment ability. Therefore, accurate prediction can reduce default risk, financial loss, and unfair credit access [2]. In recent times, machine learning (ML) and artificial intelligence (AI) have improved credit risk assessment. ML models can identify default risk from complex and heterogeneous datasets more effectively than many conventional methods [3]. Moreover, ensemble learning and feature selection can combine multiple models and identify important risk-related features, which improves model reliability [4]. XAI also supports transparency and regulatory trust. As a result, it can help build fair and explainable credit decision systems [5]. In this study, our contributions are as follows:

- A comprehensive loan approval prediction framework is developed by integrating preprocessing, Disagreement-Aware Feature Selection (DAFS), stacking ensemble learning, comparative performance evaluation, external validation, and XAI.
- We propose a novel DAFS method that combines embedded stability selection, permutation importance, and genetic-algorithm search to identify a compact and robust subset of predictive features.
- An empirical evaluation is conducted using baseline and stacking models under 10-fold stratified cross-validation, held-out testing, and external validation was also performed with two datasets, while an ablation study with DAFS and ROS verifies the contribution of each component.
- Interpretability is strengthened through SHAP, LIME, and permutation importance, and the overall framework demonstrates better performance, ROS transparency, and generalization than earlier studies [6], [7], [8], [9], [10], [11].

II. LITERATURE REVIEW

Recent research on loan approval and loan default prediction shows steady gains from ensemble learning. However, many studies still report lower performance than tuned ensemble models. They also provide limited validation, feature selection, HP tuning, and explainability support. Hota et al. [6] evaluated baseline and ensemble classifiers using

614 loan applications. Random Forest performed best, with 0.88 accuracy and 0.87 AUC. However, the study gave limited attention to explainability for transparent lending decisions. Ko et al. [7] studied AI and statistical methods for P2P loan default prediction using 60,993 records. LightGBM achieved 0.6857 accuracy and 0.7492 AUC. These moderate results show the need for stronger feature engineering, tuning, and validation. Uddin et al. [8] proposed an ensemble model using 806 samples with 13 features. The Voting Ensemble achieved 0.8726 accuracy. However, ROC–AUC, cross-validation, HP tuning, feature selection, and XAI techniques were not reported. Iftikhar and Ayaz [9] used 20,000 records with 31 features. Decision Tree and KNN achieved the highest accuracy of 86.53%. However, detailed validation, ROC–AUC, tuning, and explainability were not discussed. Reddy et al. [10] used 10,128 samples for loan eligibility prediction. Random Forest achieved 0.9188 accuracy and 0.70 AUC. Still, the cross-validation method, tuning strategy, feature selection, and XAI method were unclear. Kashyap et al. [11] used 614 records and reported 0.9071 accuracy with Linear Discriminant Analysis. However, ROC–AUC, cross-validation, and XAI were not included. To address these gaps, this study proposes an end-to-end lending decision framework with hybrid feature selection, 10-fold StratifiedKFold cross-validation, randomized HP search, tuned stacking, and SHAP and LIME-based interpretation. Final evaluation on a separate test set supports accurate and transparent lending decision-making.

III. METHODOLOGY

As illustrated in **Fig. 1**, the study begins with a loan-related dataset. First, data cleaning and consistency checks detect missing values and duplicate records. Next, categorical features are converted into numerical form through label encoding. Numerical features are then normalized using Min–Max scaling to ensure a consistent input space. After that, DAFS combines embedded selection, permutation importance, and genetic algorithm search to create a short feature subset. Finally, after stratified splitting, ROS addresses imbalance using only the training data fairly.

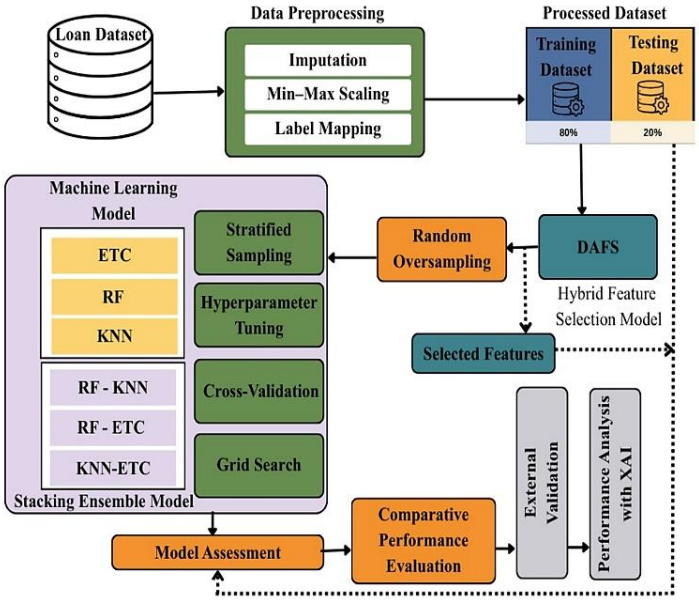


Fig. 1 Proposed DAFS-based loan approval methodology.

A. DATASET DESCRIPTION

The dataset used in this study is a synthetic loan-approval classification dataset, containing 45,000 loan records [12]. Each record includes 14 predictive features, spanning mixed data types: six continuous numeric features, which represent quantitative values such as income or loan-related measures; three discrete numeric variables, which represent countable or ordinal values; and five categorical variables, which describe characteristics of the applicant or loan. The target column, 'loan_status', is a binary variable, where 1 denotes an approved application and 0 denotes a rejected application. This structure supports supervised learning for loan-approval modeling and allows for evaluation using standard binary-classification metrics and validation protocols.

B. DATA PREPROCESSING

Before model training, the dataset is checked for consistency and suitability. First, categorical variables are converted using label encoding. Next, numerical features are normalized with Min–Max scaling for comparable ranges. Z-score standardization then reduces magnitude effects in scale-sensitive steps. After that, the data are split into stratified 80/20 training and test sets to preserve class distribution. Random oversampling is applied only to the training set to address class imbalance. To reduce redundancy and improve generalization, this study proposes Disagreement-Aware Feature Selection (DAFS). **Algorithm 1** summarizes the method. DAFS computes three feature-importance scores and combines them into one final ranking. First, an embedded stability score e_j is obtained by repeatedly fitting an Elastic Net logistic model on bootstrap subsamples. This measures how consistently each feature receives a nonzero coefficient, following stability selection [13], as defined in **Eq. (1)**.

$$e_j = \frac{1}{B} \left| \left\{ b : \left| \beta_j^{(b)} \right| > 0 \right\} \right| \quad (1)$$

Here, e_j is the embedded stability score of features j, B is the number of bootstrap fits, $\beta_j^{(b)}$ is the Elastic Net coefficient of feature j in bootstrap run b. Second, a permutation-based importance score p_j is computed as the performance drop when a feature is randomly permuted, and these permutation scores are aggregated across multiple candidate classifiers using weights based on cross validation performance [14], as expressed in **Eq. (2).**

$$p_j = \text{Score}(X, y) - \text{Score}\left(X^{\text{perm}(j)}, y\right) \quad (2)$$

Here, p_j is the permutation based importance of feature j, Score$(\cdot)$ is the chosen evaluation metric-computed on a validation set, and $X^{\text{perm}(j)}$ denotes the same data matrix after randomly permuting the values of feature j. Third, a wrapper-based GA score g_j is produced by a genetic algorithm that searches feature subsets using cross validated performance with a sparsity penalty; the final $g_j$reflects how frequently a feature appears in the best evolved subsets [15], as given in **Eq. (3)**.

$$g_j = \frac{1}{|\mathcal{E}|} \sum_{z \in \mathcal{E}} z_j \quad (3)$$

Here, g_j is the wrapper based GA score of features j, $\mathcal{E}$i s the set of elite feature subsets returned by the genetic

algorithm, $|\mathcal{E}|$ is the number of elite subsets, and $z_j \in \{0,1\}$ indicates whether feature j is included in subset z. Unlike common hybrid feature selection that simply averages scores or uses voting, DAFS explicitly reduces unstable features by penalizing disagreement among the three selectors. The novel fusion equation used in this study is given in **Eq. (4)**.

$$s_j = \sigma(w_e e_j + w_p p_j + w_g g_j - \eta \,\mathrm{Var}(e_j, p_j, g_j)) \quad (4)$$

where s_j is the final importance score of feature j e_j, p_j, and g_j are the normalized importance scores obtained from the three selectors, w_e, w_p, and w_g are nonnegative weighting, $\mathrm{Var}(e_j, p_j, g_j)$ measures disagreement among the selectors for feature j, η controls how strongly disagreement is penalized and $\sigma(\cdot)$ is the sigmoid mapping to keep scores bounded.

Algorithm 1: Disagreement-Aware Feature Selection (DAFS)

Input: $X \in \mathbb{R}^{n \times d}$, y
Output: $F_{\text{final}} \subseteq \{1, \ldots, d\}$
Parameters: B, K, G, $k_{\min}$, $k_{\max}$, η, κ

1. Compute Elastic Net stability scores e_j from B bootstrap resamples.
2. Compute cross-validated permutation importance scores p_j using K-fold validation.
3. Compute GA-based inclusion scores g_j from elite subsets over G generations.
4. Normalize e_j, p_j and g_j to the range $[0, 1]$.
5. Fuse the normalized scores into a disagreement-aware score s_j for each feature.
6. Derive the adaptive cutoff τ from the score distribution using κ.
7. Form the candidate set $F = \{ j \mid s_j \geq \tau \}$.
8. If $|F| < k_{\min}$, retain the top $k_{\min}$ features ranked by s_j.
9. If $|F| > k_{\max}$, retain the top $k_{\max}$ features ranked by s_j.
10. Return the retained set as F_{final}.

After applying DAFS, the 13 original predictors were reduced to 8 selected features: Person Income, Person Home Ownership, Loan Amount, Loan Interest, Loan Interest Rate, Loan Percent Income, Credit Score, and Previous Loan Defaults On File. This design is useful because it prioritizes features that are consistently supported by different selectors. Thus, it improves robustness across data splits and reduces overfitting compared with simple averaging or voting. Here, the disagreement term, $\mathrm{Var}(e_j, p_j, g_j)$, controls stability. If one selector gives extra weight without support from others, variance increases and s_j decreases. In contrast, balanced selector weights lower variance and provide a more reliable input space for classifiers.

C. RF–KNN Model Construction

The RF–KNN model is built as a stacking ensemble that combines two complementary learning mechanisms for loan approval prediction [16]. At the base level, KNN and RF are used as first-stage classifiers. KNN captures local similarity patterns by comparing an applicant with nearby samples in the feature space. This is useful when approval behavior depends on the closeness of applicant profiles. In contrast, RF captures nonlinear relationships through multiple decision trees. Therefore, it can handle complex decision boundaries. Instead of depending on one learner, the prediction outputs of KNN and RF are passed to a Logistic Regression meta-classifier [17]. This meta-learner combines the decision patterns of both base models and produces the final prediction. In this design, pass-through was disabled. Thus, the meta-learner used only base-model outputs to focus on learned decision behavior. To ensure robustness, the ensemble was trained under nested cross-validation. A 10-fold stratified outer loop estimated performance. Meanwhile, a 3-fold randomized HP search tuned the model. The search selected a distance-weighted KNN with a small neighborhood and Manhattan distance behavior. It also selected an RF with 50 trees, maximum depth of 18, controlled leaf constraints, and no bootstrap sampling. In addition, Logistic Regression used an L2-regularized configuration to stabilize the final combination. After fold-based selection, the best RF–KNN model was retrained on the full training set and evaluated on the held-out test set. Overall, this design combines local reasoning, tree-based ensemble learning, and nested tuning to improve robustness, reduce overfitting, and support reliable prediction.

IV. Results And Discussion

A 10-fold cross-validation protocol was applied to evaluate the RF–KNN stacking ensemble, where the Logistic Regression meta-learner was used, to obtain a reliable estimate of the model performance. The results for each fold, along with the mean value and standard deviation, are presented in **Table I**.

TABLE I. 10-FOLD CV RESULTS

	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Fold 1	0.9681	0.9685	0.9681	0.9681	0.9954
Fold 2	0.9696	0.9699	0.9696	0.9696	0.9951
Fold 3	0.9634	0.9638	0.9634	0.9634	0.9937
Fold 4	0.9683	0.9686	0.9683	0.9683	0.9960
Fold 5	0.9665	0.9671	0.9665	0.9665	0.9954
Fold 6	0.9676	0.9678	0.9676	0.9676	0.9958
Fold 7	0.9676	0.9681	0.9676	0.9676	0.9956
Fold 8	0.9658	0.9667	0.9658	0.9658	0.9955
Fold 9	0.9705	0.9707	0.9705	0.9705	0.9953
Fold 10	0.9643	0.9646	0.9643	0.9643	0.9933
Mean	0.9672	0.9676	0.9672	0.9672	0.9951
Standard Deviation	0.0021	0.0020	0.0021	0.0021	0.0009

The model achieved 0.9672 ± 0.0021 accuracy and 0.9951 ± 0.0009 ROC–AUC, with very little variation among the folds. HP optimization converged to a concise and consistent configuration, which is presented in **Table II**. In the case of KNN, the selected setting uses distance-based weighting with 3 neighbors, which implies that nearby samples have a greater impact on the prediction. For Random Forest, a maximum depth of 18 and 50 trees are selected, which ensures robust performance while controlling complexity.

TABLE II. FINAL HYPERPARAMETER CONFIGURATION

Component	Hyperparameter	Best value
KNN	weights	distance
	p	1
	n_neighbors	3
	leaf_size	20
	algorithm	ball_tree
RF	n_estimators	50
	min_samples_split	2
	min_samples_leaf	2
	max_depth	18
	bootstrap	False
Meta-learner (LR)	C	0.1
	penalty	l2
	solver	saga

Logistic Regression with L2 regularization is finalized as the meta-learner, where C = 0.1 and solver = saga are used. This ensures stable adaptation of learners, reduced overfitting, improved generalization and predictive stability. Using the selected hyperparameters, the final stacking model is trained on the training set and evaluated on the held-out test set. As reported in **Table III**, the model achieves a test accuracy of 0.9693 with an ROC–AUC of 0.9958. **Fig. 2** shows the confusion matrix on the held-out test set, highlighting the distribution of correct and incorrect predictions across the two classes

TABLE III. TEST-SET PERFORMANCE (RF–KNN Ensemble Stacking)

Accuracy	Precision	Recall	F1-Score	ROC–AUC
0.9693	0.9696	0.9693	0.9693	0.9958

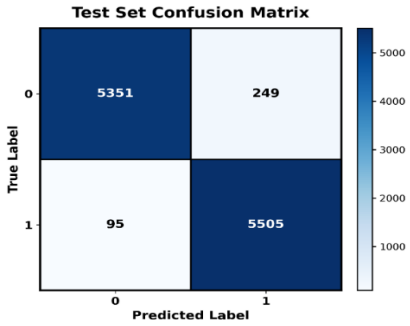


Fig. 2 Confusion matrix of the RF–KNN ensemble stacking model.

A. Comparative Performance Evaluation

All models follow the same experimental protocol for fair comparison. **Table IV** reports held-out test results for baselines and stacking ensembles. The RF–KNN stacking ensemble with a Logistic Regression meta-learner achieves the highest overall performance. Meanwhile, individual classifiers score lower under the same setting. As **Fig. 3** shows, the ROC curve indicates better class separability than the baseline models.

TABLE IV. TEST-SET PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
RF	0.9614	0.9624	0.9614	0.9614	0.9951
KNN	0.9362	0.9413	0.9362	0.9360	0.9665
ETC	0.9305	0.9351	0.9305	0.9304	0.9878
RF + ETC	0.9663	0.9665	0.9663	0.9663	0.9951
KNN + ETC	0.9560	0.9571	0.9560	0.9560	0.9915
RF + KNN	**0.9693**	**0.9696**	**0.9693**	**0.9693**	**0.9958**

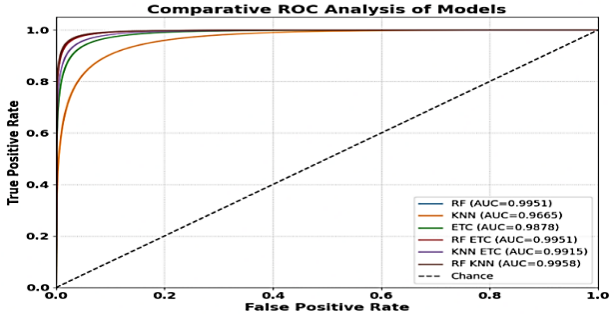


Fig. 3 ROC curve comparison of the evaluated models.

The impact of DAFS and ROS is examined by switching each component on and off, as shown in **Table V**. Applying DAFS without ROS decreases performance, showing that feature selection alone is insufficient when class imbalance remains unresolved. Applying ROS without DAFS also fails to improve performance, showing that resampling alone is not the key component. The best results occur when DAFS and ROS are used together, confirming their joint effectiveness for reliable predictions.

TABLE V. EFFECT OF DAFS AND ROS

Combination	Accuracy	Precision	Recall	F1-Score	ROC–AUC
DAFS + No ROS	0.7778	0.6049	0.7778	0.6806	0.6886
No DAFS + No ROS	0.9270	0.9255	0.9270	0.9252	0.9736
No DAFS + ROS	0.9231	0.9221	0.9231	0.9201	0.9696
DAFS + ROS	**0.9693**	**0.9696**	**0.9693**	**0.9693**	**0.9958**

B. External Validation

To evaluate the generalization ability of the proposed framework beyond the primary dataset, external validation was performed using two independent public loan approval prediction datasets taken from Kaggle **[18]**, **[19]**, as summarized in **Table VI**. Although the two external datasets differ in terms of sample size and feature composition, the model consistently maintained strong predictive performance, achieving accuracies of 0.9836 and 0.9875, respectively. The stability of these results across both external datasets indicates that the learned decision patterns are not limited to a single data distribution and can be effectively transferred to unseen loan approval data.

TABLE VI. EXTERNAL VALIDATION RESULTS

Source	Shape	Accuracy	Precision	Recall	F1-score	ROC-AUC
[18]	4269 × 13	0.9836	0.9837	0.9836	0.9836	0.9962
[19]	3,192 × 14	0.9875	0.9878	0.9875	0.9875	0.9963

C. XAI-Based Model Interpretation

To make the model's decision more transparent and support its interpretability, global interpretation using SHAP and local interpretation using LIME were applied to the combined test features **[20]**. **Fig. 4**, presents the SHAP summary, showing the overall influence and value-based direction of each feature on the model output.

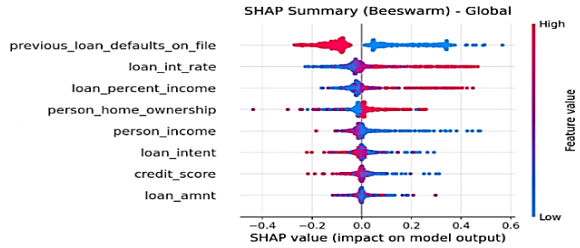


Fig. 4 SHAP summary (beeswarm) for global feature impact.

The widest spread suggests that 'previous_loan_defaults_on_file' is the most influential, followed by 'loan_int_rate' and 'loan_percent_income'. Logically, **Fig. 5** shows the LIME interpretation for a test sample, where favorable 'previous_loan_defaults_on_file' contributes positively, while more 'loan_int_rate', less favorable 'loan_intent', and more 'loan_percent_income' contribute negatively. **Fig. 6** shows the permutation feature importance, which more strongly confirms the same main drivers identified by SHAP. Overall, these explanations show that decisions are primarily driven by financially meaningful risk factors **[21]**.

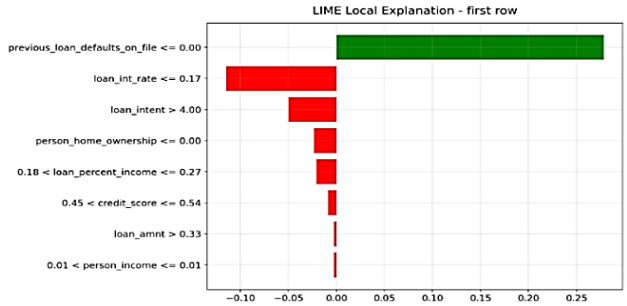


Fig. 5 LIME local explanation for a representative test instance.

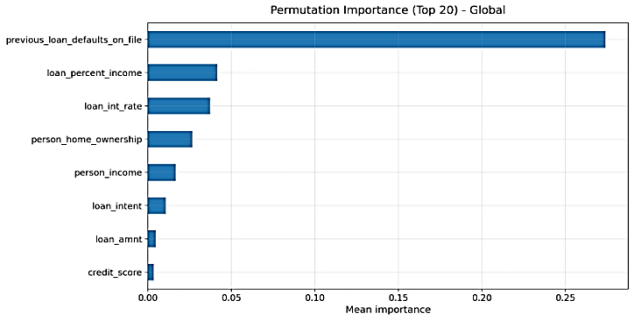


Fig. 6 Permutation feature importance for global sensitivity analysis.

D. Comparison with Other Related Works

As shown in **Table VII**, the proposed RF-KNN model achieved 96.93% accuracy, which provided better results than the existing related studies. This result highlights the effectiveness of the adopted preprocessing technique to improve the classification performance. In addition, the selected features worked together efficiently, which increased the robustness of the model. Therefore, these results verify that the proposed method is effective, robust, and competitive with previous works for practical applications.

TABLE VII: PROPOSED MODEL COMPARISON WITH OTHER RELATED WORKS

Existing Literature	Dataset Size	Methods	Acc	ROC-AUC	CV	XAI
Hota et al. [6]	614	RF + PCA-based feature selection	0.8800	0.8700	✗	✗
Ko et al. [7]	60,993	LightGBM	0.6857	0.7492	✗	✗
Uddin et al. [8]	806	Voting Ensemble (RF, ET, KNN)	0.8726	-	✗	✗
Iftikhar and Ayaz [9]	20,000	DT	0.8653	-	✗	✗
Reddy et al. [10]	10,128	RF	0.9188	0.7000	✗	✗
Kashyap et al. [11]	614	RF with LDA dimensionality reduction	0.9071	-	✗	✗
Proposed Model	**45,000**	**RF–KNN with DAFS**	**0.9693**	**0.9958**	✓	✓

V. Conclusion

This study presents a comprehensive framework for loan approval prediction, which combines DAFS-based feature selection, stacking ensemble learning, external validation, and XAI. It shows that combining DAFS with explainable stacking ensembles enables reliable loan approval prediction while maintaining transparency of the decision-making process. The proposed method achieved an accuracy of 0.9693 on a separate test set, and 0.9836 and 0.9875 on two independent public datasets, respectively, indicating its applicability beyond the main dataset. The model is explained using SHAP, LIME, and permutation importance at both the global and instance levels, which increases transparency. However, the study was validated on a limited number of public benchmarks, and real operational environments may have wider distribution shifts and additional constraints. Future studies will examine more diverse, time-separated datasets and include calibration and application-specific threshold selection for practical applications.

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